

Supporting multiple runs per vehicle in a routing plan

Last updated on Oct 29, 2024

Context

Currently, whenever a vehicle is selected at the time of creating a new route plan in the Route Optimizer FPA, only one route is created for that vehicle in that plan. However, there may be scenarios where the tenant would want to utilize the same vehicle for multiple consecutive routes on the same day. In this case, all orders for the day are available/ created upfront for route-planning

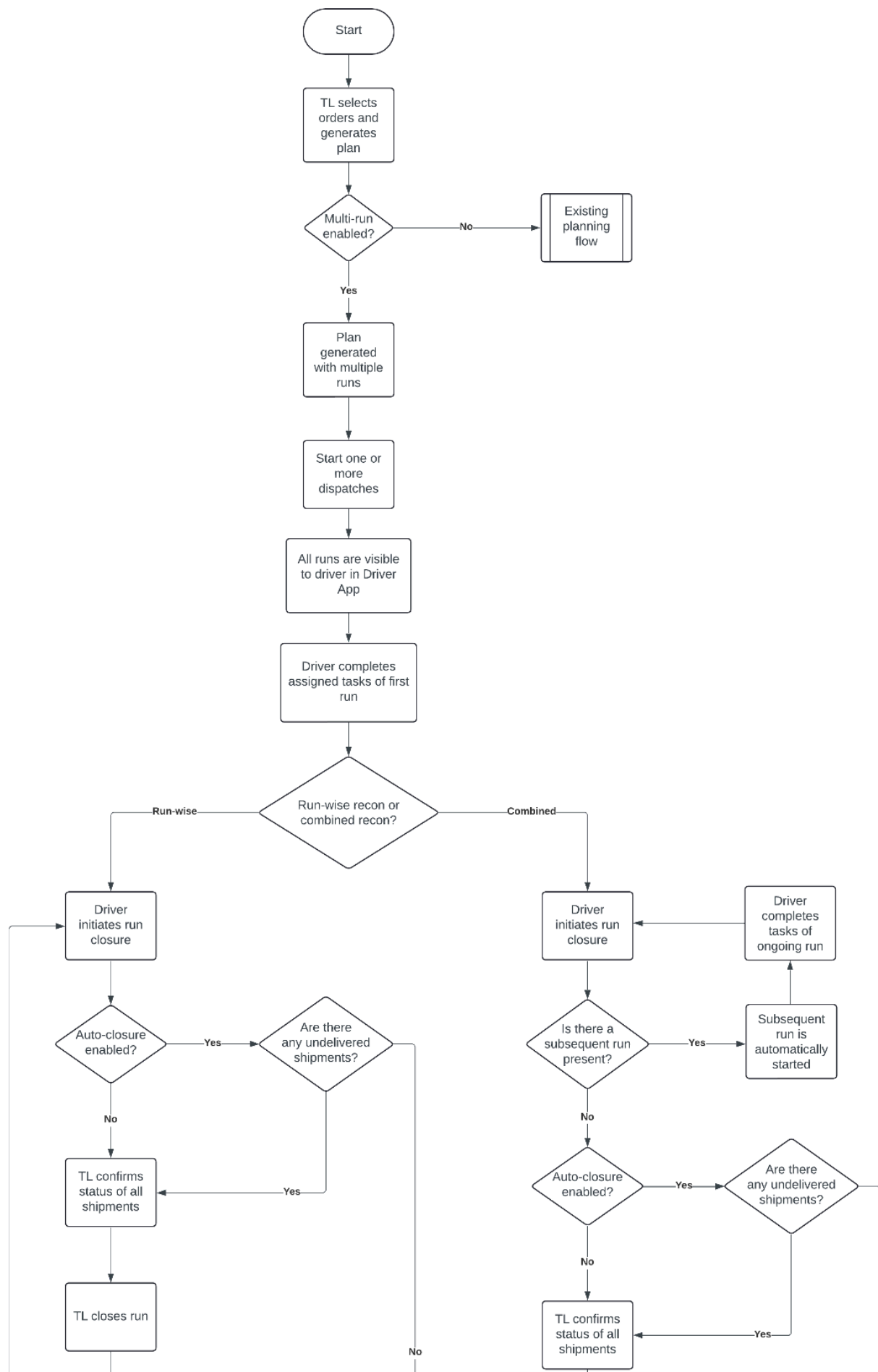
For example, IOCL distributors use 3-wheelers to deliver LPG cylinders to retail customers. However, due to the limited capacity of these vehicles, the same vehicle needs to come back to the pickup facility (distributor warehouse) multiple times in a day to fulfill all its orders.

This document captures the changes required on DispatchOne to enable support for drivers performing multiple runs [\[Applicable only for SPMD and MPSD use cases \]](#)

Definitions

1. **Run:** Each circuit within a given route plan that a vehicle completes from base facility → pickup/drop locations → base facility. A vehicle can have multiple runs within a plan.
2. **Plan:** An instance of the “Plan” entity is created every time a user selects orders and proceeds with the option to “Create New Plan”. Such a Plan will persist either until it is discarded, or until all its Routes are started and there are no remaining unassigned orders.

Proposed Workflow



Designs

1. [Routing](#)
2. [Dispatch](#)
3. [FE App](#)
4. [Orders Map View](#)

Proposed Approach

System Design

1. A tenant-level configuration should be introduced to identify if the custody handover step for a run should be performed at the end of each run, or should be performed together for all runs belonging to a driver.

Plan Creation

1. In the Plan Creation modal, a configuration titled “enable multiple runs?” should be introduced against that plan [\[vehicle-wise config not required\]](#) under Advanced Configurations. If this is enabled then any number of runs should be allowed against each vehicle.
2. If a TL has enabled multiple runs, then the route optimizer output will create each run of such a vehicle as a separate route. Whenever a driver has multiple runs in a plan, its corresponding runs should be grouped together under a single route in the “Routes” table in the Plan Details page. Driver and vehicle assignment will be against a group and each run of a group will have the same driver/vehicle combination.
3. These grouped runs should be listed in the sequence in which they are to be dispatched and should have the serial number displayed against each run
4. A TL should be able to discard an individual run even if it belongs to a group. Doing so should move its orders to the “Unassigned Orders” section of the plan.

Dispatch Creation

1. When a plan is published, the first run will move to “Ongoing” state and the driver will be able to act on its orders, while the subsequent runs will be in “pending publish” state.
2. TL should not be allowed to make any changes to subsequent runs which are in “pending publish” state.

3. When the “Assign and Start” button is clicked on a plan, all the first runs should get published and move to “ongoing” state, while subsequent runs should move to “Pending Publish” state.
4. There will be no option for a TL to selectively publish/start runs from within a group of runs assigned to a driver and vehicle combination. Only the group itself can be published, not individual runs inside it. This is because if only a single run of a driver is published, then multiple complex route-level validations need to be introduced at the time of replanning, comparing, and at the time of publishing the remaining runs to ensure that they do not overlap with the slots of the already-published run.
5. In the Dispatch List page, all the runs belonging to a driver should be grouped and displayed together. “Groups” are valid only for runs which are part of the same plan and have been published together, and not across plans. This is because of the following reasons:
 - a. For a given plan, we allow the TL to add new orders to it any time and replan it. Such additional orders may include those which have slots overlapping with an existing run of a driver.
 - b. For runs created for the same driver across two different plans, there is no way to ensure that the orders do not have overlapping slots.
6. For runs in “Pending Publish” state, the assigned drivers and vehicles should be displayed against them on the UI. The assigned driver/vehicle cannot be changed at this stage.
7. TL should also be able to remove orders from any upcoming run until it is started.
8. A run in “Pending Publish” state cannot be started until the preceding run has been marked **closed** by the Driver from the Driver App or if the TL has completed the “End of Run” flow from web.
9. Whenever one run of a driver is closed, then the subsequent run should automatically be started. This will repeat until all runs are closed.
10. The Dispatch runsheet csv format should be updated to include a “Trip” column (format linked [here](#))

Run and Dispatch Closure

1. A tenant-level flag should be created to identify if the customer needs to complete inventory and cash reconciliation against each run, or in one go at the end of all runs.

Driver App

1. When a plan is published, or when a set of routes belonging to the same group are published, then all the runs should be pushed to the driver and the details of these runs should be made visible to them on the Driver App.
2. These runs will include the first run along with the subsequent runs
3. The Driver will be displayed a summary of each run, including the following information:
 - a. # of orders
 - b. Duration of run
4. In the task list page (home screen), the driver should **only be able to act on the tasks of the first run**.
5. Tasks of subsequent runs will be displayed as and when those runs are started.

Order FPA

1. In the Order Details page > Order History widget, the Trip number should additionally be displayed alongside wherever the Dispatch ID is displayed
2. In the Order Maps page, the dispatch cards should now display constituent trips too
3. In the Order Maps page, when the user selects unassigned orders to add to a dispatch, only the “ongoing” trips should be displayed on the map (for dispatches having multiple trips)
4. The “Add to Dispatch” modal from the Order List page has been updated with a new UI which shows Driver, Vehicle, and Trip No.

Data Storage

1. Group ID should be captured against each run and stored in DWH so that analytics can be generated at both run- and group-level.

Future scope

The following features are left out of scope in the phase 1 implementation of multi-run, and will be added incrementally (in the same sequence as below):

1. Removing orders from an ongoing run
2. Adding orders to an upcoming run
3. Adding run or group-level metrics on real-time and historical dashboards
4. Creating multiple runs through manual dispatch creation

5. Route approval:
 - a. As part of a separate requirement, we will also eventually be building a feature for route approval, wherein individual routes in a plan can be sent to the concerned Driver or Vendor for approval.
 - b. Whenever the route approval feature is introduced, if a driver has multiple runs in a plan, all routes (each route representing one run) need to be sent for approval together.
 - c. Similarly, the Driver or Vendor will have to approve or reject all the routes in a group together.

Feature Flag Combinations

When multi-run is enabled and configured for a tenant, care should be taken to ensure that it does not conflict with any of the existing feature flags.

1. Multi-run should not be enabled for a tenant where the vehicle odometer reading feature is enabled. The flows described in this document **do not** include manually capturing vehicle odometer readings (from either the Driver or the TL), as this results in too many steps for the users to perform.
2. If multi-run is enabled for a tenant and configured such that the EOD happens in a combined manner at the end of all runs, then the *auto_close_dispatch* flag should mandatorily be set to FALSE. The flag *auto_close_dispatch* determines whether the Driver can unilaterally end a dispatch and perform custody transfer without any intervention from the TL. By setting it as FALSE, TL intervention would be mandated, which is logically inconsistent with combined EOD.

Discussions

Nov 5, 2024

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- Routing:
 - Route details page to be added
 - Prabhat Raju to add to Figma | Closed ▾
 - Rider/vehicle edit flow needs to be added
 - Prabhat Raju to add to Figma | Closed ▾
 - Multi-run with return to base disabled
 - No change in UI, Shivam Manchanda to check and get back if this is possible using LogisticsOS API. Expectation is that if return to base is disabled, the last stop of the last run should be a customer address (based on optimized sequence) and should not be the base facility. The last stop of each of the preceding runs will continue to be the base facility.
Pending ▾
 - Dispatch:
 - Will we enable multi-run for manual dispatches in the future? What will this look like (high-level)?
 - No, multi-run will always be only for dispatches created through routing Closed ▾
 - In the Dispatch List page, by what logic should the dispatches be listed under the various tabs/statuses?
 - Dispatch state should transition along with its last constituent run Closed ▾
 - What is the impact on the Dispatch Details page with eod? (impact on other screens for new flows)
 - Prabhat Raju to add to Figma | Closed ▾
 - Need to add odometer flow + mts distance flows in designs
 - We will keep odometer reading flow separate from multi-run, and build for it later if/when we get any actual use cases from customers Closed ▾
 - Should we allow the addition of orders after publish?
 - Yes, doc updated Closed ▾
 - What will the flow for partial order addition look like when orders are added after dispatch publish?
 - Prabhat Raju to add to Figma | Closed ▾
 - How to handle dispatch-level validations if one run ends and the next run publish fails? Eg: e-waybill, partially scanned MPS orders, etc.
 - We will keep ewaybill flow separate from multi-run, and build for it later if/when we get any actual use cases from customers Closed ▾

- Call out wherever calculation of metrics will change as a result of multi-run Closed ▾
 - Route Plan Details page:
 - Vehicles used count should be at a group-level. If a vehicle will be used for 5 runs under 1 group it should only be counted once in the metric card above
 - Should undelivered orders carry over to subsequent runs?
 - No. Not on the roadmap either. Closed ▾
 - If closure of an entire group of runs is initiated, should orders from previous closed runs appear in the EOD step?
 - Yes Pending ▾

Rough Work -- Do not read further

Sample User Flow

1. TL selects orders and chooses to create a new route plan
2. TL chooses if they want to enable multiple runs or not against ALL vehicles in that plan
3. TL enters remaining constraints and creates the plan
4. After examining the plan details, TL clicks on “Assign and Start Routes” for the entire plan
5. All the runs get created under a single dispatch. The first runs move to “ongoing” state, while all the subsequent runs get created in “Pending Publish” state. [Subsequent runs are part of the dispatch and are not an independent dispatch]

6. Consider a Vehicle A which has two runs in the plan. Vehicle A completes its orders of the first run and the Driver marks the run as completed from the Driver App.
7. TL completes the EOD process for the first run of Vehicle A by confirming handover of inventory and cash
8. On the Dispatch List page, the TL is notified that the second run for Vehicle A is now available for publishing [\[This can be removed for tenants who don't want any reconciliation in system at the end of 1st run\]](#)
9. TL starts the second run for Vehicle A [\[Will be automatic depending upon #8\]](#)
10. Vehicle A completes its orders and the Driver marks the run as completed from the Driver App.
11. TL completes the EOD process for the second run of Vehicle A by confirming handover of inventory and cash
12. All runs of Vehicle A are now complete

Life cycle of a run/trip:

- Assigned: When the dispatch to which the run belongs is in “Assigned” state
- Ongoing: When the run has been started by the TL or Driver
- Upcoming: When the dispatch to which the run belongs has been started but the trip itself has not yet started
- Pending Closure: The dispatch is ready for any required reconciliation and custody transfer.
- Closed: The custody ledger of the dispatch is reconciled and the dispatch has been completed.

